

Hook & Match - Development and Evaluation of a Pair Matching Phishing Card Game

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Digital fatigue is increasing due to ongoing digitization, yet understanding digital processes remains crucial. Cybercriminals exploit the lack of knowledge and awareness among users. To address this, we developed “Hook & Match”, a Pair Matching Phishing Card game, which combines the familiar non-digital principle of pair matching with phishing education to avoid contributing to digital fatigue. In “Hook & Match”, players match cards to learn how to distinguish different phishing attempts from legitimate examples.

Our evaluation shows that playing “Hook & Match” significantly improves users’ ability to identify phishing compared with legitimate cases. We also investigated whether prior knowledge had an impact on the learning outcomes and found that, regardless of their initial level of expertise, both knowledgeable users and novices benefited from playing the game. However, participants with the lowest initial level benefited the most. This demonstrates that “Hook & Match” is an effective tool for phishing education without exacerbating digital fatigue.

CCS Concepts: • **Security and privacy** → **Usability in security and privacy**; **Social aspects of security and privacy**.

Additional Key Words and Phrases: Security, artifact or system, quantitative methods, user study, SeBIS, SA-6, phishing

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1 Introduction

As cyber attackers continue to exploit human vulnerabilities, phishing attacks have become one of the most pervasive and insidious threats in the digital landscape. The **Anti-Phishing Working Group (APWG)** reported a record number of 4.7 million phishing attacks in 2022, representing an alarming increase of over 150% for each year since 2019 [1]. These statistics underscore the critical need for effective countermeasures to prevent and mitigate phishing attacks.

Phishing awareness measures are already a large area of focus, with significant improvements in people’s ability to distinguish phishing from legitimate messages achievable through targeted

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training. An approach that has shown promise is game-based learning, which leverages interactive and engaging experiences to educate users about cybersecurity best practices [2–4]. However, most existing measures have been designed digitally, mainly because they relate to a digital topic.

Moreover, increasing digitization of all areas of life has led to a growing concern: digital fatigue. As people become increasingly overwhelmed by the demands of the digital world, their ability to cope effectively with cybersecurity threats is compromised [5]. This phenomenon highlights the need for alternative approaches that do not contribute to digital overload.

In addition, most measures are designed under the concept that someone learns by being taught, rather than through self-directed learning or active problem-solving. Despite this, research suggests that active learning, in which individuals engage in hands-on problem-solving and critical thinking, can significantly improve the effectiveness of learning [6, 7]. By allowing learners to discover concepts and solutions on their own, they are more likely to internalize knowledge and apply it in real-world scenarios [8].

Our study addresses these problems by evaluating an analog game-based approach to phishing awareness training. The “Hook & Match” is designed to increase the user’s competence in distinguishing between phishing and legitimate messages through interactive gameplay, without relying on digital technology. In contrast to existing measures, our approach provides a unique opportunity for users to engage with cybersecurity education in a non-digital context.

In this article, we present four key contributions:

- (1) Development of an analog single player self-directed learning game for the digital topic of phishing
- (2) The game receives positive feedback and seems to be enjoyable for players
- (3) The game contributes to a more effective distinction between phishing and legitimate messages
- (4) Initial insights into the relationship between general security behavior intention/attitude and improvement in distinguishing between phishing and legitimate messages

2 Related Work

Research in security awareness has led to various measures being developed, including games that focus on game-based learning compared with gamification, where only gaming elements are added to an intervention [9]. Game-based learning aims to maintain motivation during gameplay and achieve better effects in terms of effectiveness.

2.1 Security Awareness Measures

several awareness-raising measures are available, although there are still some gaps [10]. These measures include videos on phishing [11] and games that address different areas of knowledge in software engineering [12] or specific topics such as phishing [13]. Most of these approaches require both the visual design and content development phases, enabling the transportation of information visually and textually. However, they often have device-specific requirements, such as laptops or AR/VR devices. Examples include games for regular computers [14–18] and apps for mobile devices [19, 20]. The closest work to our game is Protect [21], which requires players to match given attacks with corresponding defenses, but it is a digital game that covers social engineering in general.

2.2 (Card) Games and Game-Based Learning

Analog tabletop games for security training or awareness raising are numerous, targeting different areas in academia [22–28]. These games are often more complex than our game. Analog card games have been used for centuries in various forms, including those that introduce functional programming concepts [29]. Anyway, many of these games require a lecture beforehand and have

only been tested with computer science students. Commercial security mini-games similar to our game often lack public descriptions and published evaluations.

Our game is closest to the work by Fatima et al. [30, 31], which uses social skills flashcards for kids, presenting scenarios on one side of the card and asking questions about whether to open links or not. Although both games share a similar purpose and focus on phishing, our game covers more topics within the domain.

2.3 Self-Directed Learning

Effective learning methods have been extensively studied in educational research, and various approaches have emerged to address different learning needs and goals. Self-directed learning, often associated with adult education and lifelong learning, emphasizes autonomy and self-regulation as crucial factors for successful knowledge acquisition. Research by Knowles [32] has highlighted the importance of self-directed learning in fostering critical thinking skills and personal growth, although it also points out that this method requires learners to have prior knowledge and intrinsic motivation. Bandura's [33] theory of self-efficacy further supports the idea that confidence in one's ability plays a significant role in effective self-directed learning.

On the other hand, instructor-led learning represents a more traditional approach, with extensive research indicating its effectiveness in guiding learners through complex subjects. This method is particularly valued for its structured environment and immediate feedback mechanisms, which are essential for clarifying doubts and ensuring comprehension. However, critics argue that instructor-led learning may lack flexibility and adaptability to individual learning styles, potentially hindering the development of self-directed learning habits [34].

Video-based learning has gained prominence with the rise of online platforms, offering a unique combination of visual, auditory, and textual elements that enhance memory retention and understanding. Studies on multimedia learning suggest that this approach can make complex information more accessible and engaging to learners [35]. Nevertheless, challenges such as information overload or lack of interactivity have been noted in passive video consumption, highlighting the need for additional interactive elements to maximize its effectiveness [36].

In summary, existing research provides valuable insight into the effectiveness of self-directed, instructor-led, and video-based learning methods. However, self-directed seems to be under-represented in security research. This study aims to contribute by developing a self-directed security awareness measure about phishing and evaluating its effectiveness.

3 Methodology

In this chapter, we provide a comprehensive description of the necessary prerequisites and steps involved in the development (see Section 3.1) and evaluation (see Section 3.2.4) of "Hook & Match".

The first subsection begins by outlining the prerequisites derived from previous research and the essential requirements that the game must fulfill. Subsequently, we detail the concrete steps of the development process, including the structural and design aspects of the game.

The second subsection focuses on the necessary steps towards evaluating our card game. This includes a description of the pertinent research questions and existing hypotheses that guide our evaluation. In addition, we address ethical considerations and conclude by outlining the data and analysis plan required to address the research questions. The final section delves into the actual implementation of the study, providing a detailed description of the study procedure, including laboratory and online test sections.

3.1 Game Development

In the following, we explain how we developed “Hook & Match” and which factors were decisive for the development. The requirements for playing the game are determined in the first step. These are derived from previous research or experience. The next step describes in detail the process of creating the game. This includes the content and the structure of the game.

3.1.1 Requirements. The game was designed based on the following requirements. For each requirement, we provide a rationale and brief description how we designed the game to fulfill them.

Req.1 - The game should be playable without the need of a screen. The game could be designed as a computer or mobile game. However, we decided to aim for a physical game. Although both formats have their benefits and limitations, the following reasons led to a physical and analog game: A digital game would have contributed to digital fatigue by further increasing the players’ screen time. Furthermore, Denning et al. [22]’s reasons apply in our case, as well: Physical games may be attractive to people who do not like computer games. They do not require hardware or digital resources, except for a table. Physical games allow to browse its components without playing. This allows even people who do not want to play the game to benefit from it. As a result, we decided to develop a board or a card game.

Req.2 - It should be reasonably easy and affordable to distribute the game to a large group of players. Possible use cases for such a card game are very diverse. It can be distributed as a small giveaway, similar to a flyer, either personally at trade fairs or by mail. As a consequence of Req.1, the decision was made between a board and a card game. A card game seems to be the better choice as it involves less material and is therefore cheaper and more sustainable to produce and distribute. Especially a larger board game is more difficult to distribute in larger quantities at trade fairs, and the production costs for any figures and for shipping are also significantly higher.

Req.3 - Playing our game should not require a particularly large investment of time. In the organizational context as well as in the private context a shorter play time results in a higher reactance to start playing at all. Furthermore, for longer games, it is difficult to pay the same attention over time, which might result in a decreasing uptake of the knowledge/awareness if the game lasts longer.

In addition, short measures have already been shown in the past to be an effective way of conveying knowledge. In the context of this study, we define short as less than 25 minutes. This time is in the lower area of the range of 25 to 90 minutes that has been found in the literature to be effective for a learning phase [37, 38].

Consequently, similar to Req.2 a card game with a simple rule set seems to be the better choice, as it involves less material to study and therefore player spend less time before the game on learning a complex rule set. In case the time for playing is limited the time to learn a complex rule set might be reducing the time playing the game and learning the important information.

Req.4 - The game should target private individuals. Our impression is that there is more lack of support for private individuals than for employees in a working environment as the latter can be more easily targeted with specific training inside the company. This also means that texts and URLs would have to be created specifically for the tools within the company that runs the measure to be effective [39]. We address this requirement by choosing examples from personal life areas such as online shopping or online banking, rather than areas more typical in the business world (e.g. mails from human resources or invitations to online meetings).

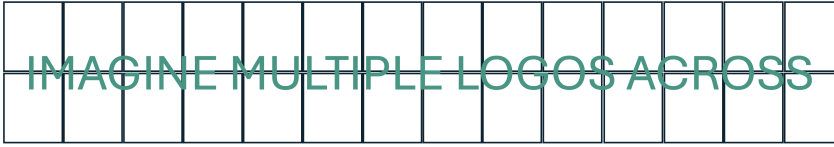


Fig. 1. Exemplary illustration of the back of the cards (anonymized) forming multiple logos as a solution for users to know what cards are at the right position.

Req.5 - The game should use a format common to the majority of players. The game should use a format that many people are familiar with. Considering Req.1 this would be either a card game or a board game. However, as already discussed, a card game seems to be the better choice. Card games represent a common form of leisure activity and have existed for decades if not centuries. As a game mode, we consider matching two cards to each other, which has also existed in many forms over a very long period of time e. g. (quizlike) Memory or Domino.

Req.6 - The game should work without the involvement of an expert or game master. The need for a game master or expert significantly restricts the ease of playing as time and location need to be arranged and the player would be dependent on planned sessions. We already mentioned that the game will be based on matching pairs in Req.5. We facilitate two ways in which players can find the correct solution. Within the game: the content of the cards is designed in such a way that only one pairing matches. Even if it is the case that some cards deal with similar topics. Thus, only one suitable pairing can be found through the overall picture of all card pairs. In addition, the sentences used help to find this pairing even if the user is unsure or if several pairings appear thematically appropriate at first. Thus, the sentence structure in the combination of the two cards can often exclude a wrong pairing due to grammatical errors. This is deliberately used to stimulate the user to take active consideration of possible errors in the pairing. After the game: We have printed a graphic on the back of the cards, which only gives a correct picture if all cards are arranged correctly (see Figure 1 for an exemplary illustration). With this solution, it is possible for the users to generate the correct solution even without a game master. Afterwards, the users can check which cards they have placed incorrectly. The principle is similar to the tutor systems originally designed for students, but also used in other serious games [40]. Being able to have a look at the solution by oneself allows the players to reflect on their own solution without having to justify it to another person.

Req.7 - The game should work for a single player. While not as difficult as scheduling an appointment with a game master or expert, the need to find teammates or opponents negates many of the benefits of a short game, as it requires more planning, e. g. by scheduling an appointment. In addition, we want our work to focus on the individual activity in learning, not on team work or collaborations. We therefore decided to aim for a single player game.

Req.8 - The game should cover all areas of knowledge about phishing. Although the game should be short, our goal is to cover all relevant areas of phishing. For the identification of all relevant areas, we rely on the concept proposed by [41] which includes different forms of phishing tricks, basic awareness of the topic and assistance for the URL. The game should include an introduction to the topic, the tasks of detection of (i) implausible, fraudulent messages, (ii) messages with dangerous links, and (iii) messages with dangerous attachments. A more detailed description can be found in Section 3.1.2.

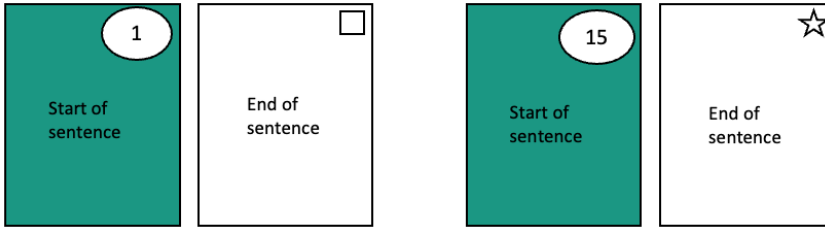


Fig. 2. Example of the design for round 1 and round 2. The card game has a number in the top right corner for the beginning of the sentence and a symbol to indicate which round the end of the sentence belongs to (rectangular or star).

In summary, we aim at creating a short paced, single player card game without supervision which is easy to learn and covers all relevant areas of phishing.

3.1.2 Game Development. For the development of “Hook & Match”, we first met the requirements listed in Section 3.1.1. Within that section, we already argue how the requirements will influence the properties of our game. Additionally, we followed guidelines and principles (e.g. interactivity increases personal agency and therefore raises the meaningfulness of a game and it’s content [42–44]) that should ensure the effectiveness of the game. The purpose of “Hook & Match” is to teach the most important tips to better distinguish between phishing and legitimate messages. The game consists of 56 different cards, played in two rounds (28 cards arranged in each round): 28 green cards with the beginning of the sentence and 28 white cards with the end of the sentence, all numbered. The goal of the game is to match the beginning and end of different sentences. Two matching cards form a correct set. The game can be played in two rounds (marked with a rectangle for round 1 or a star for round 2). Each round contains 14 sets of arranged cards (see Figure 2).

The participants start with the first set and arrange all 14 green cards (see Table 8) next to each other in ascending order and then relate the matching white cards (see Figure 3). The goal of the game is to find the correct sentence ending for each sentence beginning, i. e., to place the matching white card below the respective green sentence beginning card. It is important that the cards are in the correct order. Once the players are done with the relation, they can turn over all cards, and if the cards are arranged correctly, a solution combination of logos will appear to allow the players to check if they have matched the cards correctly. If not, players can rearrange the white cards to match the logos. Afterwards, the players can look again at the rearranged front sides of the cards to study the correct solution. The second sets of cards (see Table 9) is played in the same way.

The content of the cards is based on different attack methods that can occur in phishing e-mails. “Hook & Match” is structured as follows (for the translated version of the cards see Table 8 and Table 9 in Appendix C):

- Sender: Players should learn/know that the sender name of a mail is freely configurable as is the who-area of the sender email address, although the later might provide some hints about the sender. (Cards 1 to 6)
- Link: Players should learn/know that on a desktop device, they should use the mouse to move over a link to display the URL, because clicking on a link can lead to an attack installing malware. (Cards 7 to 10)
- HTTPS: Players should learn/know the meaning of https, but that it does not prevent criminals from also using it. (Cards 11 to 12)
- Who-Area: Players should learn/know to pay special attention to the Who-Area when viewing a URL. (Cards 13 to 14)

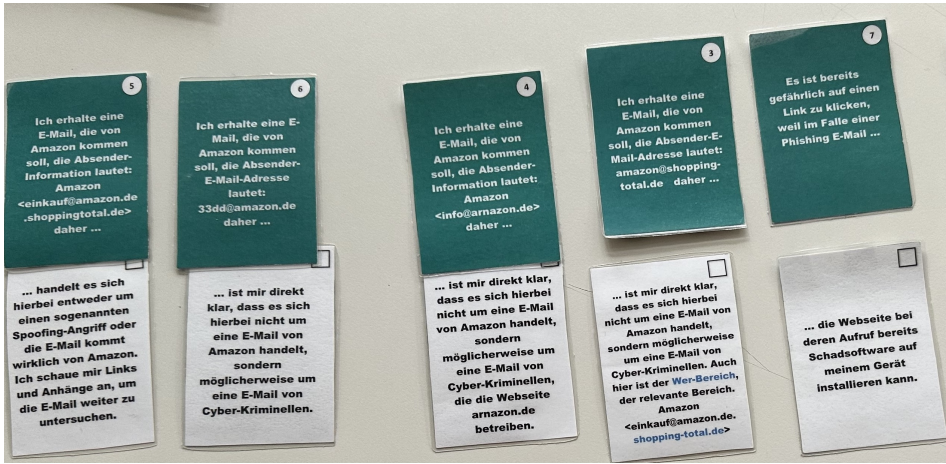


Fig. 3. Example of participant round 1 result.

URL: Players should distinguish legitimate from fraudulent URLs. Each fraudulent URL uses a different method to trick the receiver. (Cards 15 to 24)

Attachment: Players should learn/know to distinguish legitimate and potentially fraudulent mails by the suffix of the filename. (Cards 25 to 28)

3.2 Research Design

In order to determine the effectiveness of the study, it was divided into three phases (pre-test, card game, and post-test). It was particularly important that these three phases were also separated in time.

This was to prevent the previous measurement from influencing the gameplay, for example due to fatigue. Similarly, this separation was intended to prevent the first two phases, the pre-test and the card-game, from negatively affecting the third phase, the post-test with the second measurement of the card game. If the second effectiveness measurement had been conducted immediately, it may have led to more negative results that could not be attributed to the game itself.

3.2.1 Research Questions. Now that the game has been developed, the next step will be to evaluate its effectiveness. Several aspects should receive particular attention.

The game should generally contribute to improving participants' ability to distinguish between phishing and legitimate messages.

This leads to the following research question (**RQ1**): Does playing our "Hook & Match" card game improve sensitivity (distinguishing phishing from legitimate emails)?

To better identify the target group for the game, additional factors should be examined. One way to categorize people is by their previous knowledge. Since the game is not designed with a game leader or as direct knowledge transfer, such as a video, prior knowledge could have an important influence on the effectiveness of the game. Using the factors of prior knowledge in phishing and prior knowledge in general IT security, it will be investigated whether certain groups benefit differently from playing the game. Therefore, it will be examined whether different factors have an impact on the rate of improvement in distinguishing between phishing and legitimate messages. It will be investigated whether the ability to distinguish between fraudulent and legitimate messages before playing the game has an effect on how well participants improve in the game. This should

clarify whether the game is more meaningful for people who do not yet possess high abilities or perhaps only for those who already have a high level of ability. The latter could be the case because there is no instructor present to answer questions or guide the participants.

This leads to the following research question (**RQ2**): Has the “Hook & Match” card game a different impact on the sensitivity depending on the user sensitivity before playing the game?

Additionally, the game was designed not only to teach knowledge but also strengthen participants’ confidence in their own abilities to distinguish between phishing and legitimate messages. Previous studies have shown that having the necessary skills does not solely determine whether the corresponding behavior is exhibited in specific situations [45–47]. Therefore, in addition to the effectiveness of the measure, it will be investigated whether participants develop a “healthy” self-awareness regarding the topic of phishing through the measure. And, also leads to the following research question (**RQ3**): Does playing the “Hook & Match” card game positively influence participants’ self-confidence in recognizing phishing emails?

Several studies have been conducted on the use of psychometric instruments for the measurement of individuals’ security-related behaviors and attitudes. Two notable examples are the **Security Behavior Intentions Scale (SeBIS)** [48, 49] and the **Self-Report Measure of End-User Security Attitudes (SA-6)** [50]. The SeBIS assesses an individual’s intentions to engage in secure behaviors, such as using strong passwords and updating software regularly. In contrast, SA-6 measures end-users’ attitudes towards security, including their perceptions of security risks and policies. Although both scales share similarities in that they aim at understanding human factors that influence security outcomes, they differ in their focus on behavioral intentions versus attitudes. The SeBIS is more action-oriented, evaluating an individual’s plans to perform specific security-related actions, whereas the SA-6 provides insight into the underlying cognitive and affective processes driving those actions. By considering both scales, researchers can gain a more comprehensive understanding of the complex relationships between human factors and security behaviors.

The inclusion of SeBIS and SA-6 in our study provides a more nuanced understanding of the human factors that influence security behaviors. By assessing both behavioral intentions (SeBIS) and attitudes towards security (SA-6), we can identify potential discrepancies between what individuals intend to do and their underlying attitudes, which may influence their actual behavior. For instance, an individual may indicate a high intention to use strong passwords (SeBIS), but hold a negative attitude towards security policies (SA-6), potentially leading to inconsistent behavior. This dual perspective allows us to better understand the complex psychological processes driving human behavior in the context of security and, ultimately, informs the design of more effective security training and awareness programs, such as our proposed game-based approach.

The idea is to enrich the study’s credibility and depth with insights into psychological construct impacts on sensitivity. This leads to the following research question (**RQ4**): Are there differences in sensitivity in relation to users’ values for self-reported SA-6 and SeBIS, both with and without playing our “Hook & Match” card game?

3.2.2 Ethical Considerations. This study was conducted in accordance with established ethical guidelines and the requirements of our university **Institutional Review Board (IRB)** and **ethical review board (ERB)**. Before participation, all participants provided their informed consent through the Sosci Survey platform, acknowledging their understanding of the nature of the study and their rights as participants. They were also informed that they could withdraw from the study at any

time without penalty or consequence. To ensure transparency and participant awareness, the first page of the survey provided a clear explanation of the purpose and procedure of the study, while the second page outlined participants' rights. For the lab study component, the participants were specifically informed about the collection of photographs of their results and note-taking during their participation. No audio or video recordings were made. The study protocol received approval from our university's IRB/ERB prior to data collection, ensuring that all aspects of the research conformed to ethical standards.

3.2.3 Evaluation. A unique feature of our game is its evaluation methodology. Unlike other games that have limited evaluations or only measure player success directly before and after playing, our study evaluates players' knowledge and attitudes two days before and two days after playing our game [51]. This allows for more comprehensive conclusions about the effectiveness of our game. Two days are particularly helpful because they prevent the stress and tension from the pre-test (more details on the evaluation phases in Section 3.2.4) from being transferred to the actual gameplay. Participants should fully concentrate on playing during that time. Similarly, the testing of the pre-test and time spend for gameplay should have less influence than the results of the post-test and rather the learning from the card game influencing the post-test results. Furthermore, positive influences of the game should not drop so significantly within two days that they can no longer be measured. Studies have shown that up to 80% of what is learned is forgotten in the first 24 hours [52]. If the game does not produce sufficiently good results even after only two days, then it indicates a problem with the game itself rather than a methodological issue in measurement. With this study design and the planned temporal distance, we also aim at preventing the learned knowledge from being retained solely in short-term or working memory, thereby also ensuring it has a chance to endure over a longer period [53].

On the other hand, good results after two days would indicate that the content of the game is much better remembered through active retrieval of information via tests or quizzes [54]. Of course, two days are not enough to make statements about long-term effects. But this study is meant to provide a first indication and if there are positive results, then appropriate long-term studies should follow, as other authors have already demonstrated [11, 55].

3.2.4 Game Evaluation Methodology. The study was divided into three phases (pre-test, card game and post-test). The pre-test and post-test phases measured the variables phishing sensitivity, security awareness and socio-demographics, respectively. Figure 4 gives a short overview of the content of all these three phases.

More details on the structure of the tests (pre-test and post-test) can be found in Section 3.2.5. The card game phase section represents the actual card game and the corresponding feedback (see Section 3.2.7). In comparison to other studies, it was decided to separate the respective sections temporally. This had the following reasons: (1) The playing of the card game should not be distorted by the pre-test and the associated "load", (2) The pre-test and post-test should not directly follow each other in time, so that a "recognition" of the examples is reduced,

(3) The results of the post-test should not be negatively influenced by an already longer study duration with pre-test and card game.

(4) The results of the post-test should not be negatively affected by a study duration that has already been prolonged with the pre-test and card game.

Accordingly, participants were sent the invitation to the pre-test two days before their appointment for the card game. Similarly, we sent participants the post-test after the laboratory appointment with an interval of two days. Thus, the entire study was carried out within a week for all participants. This does not give any indication of a long-term effect of what has been learned,

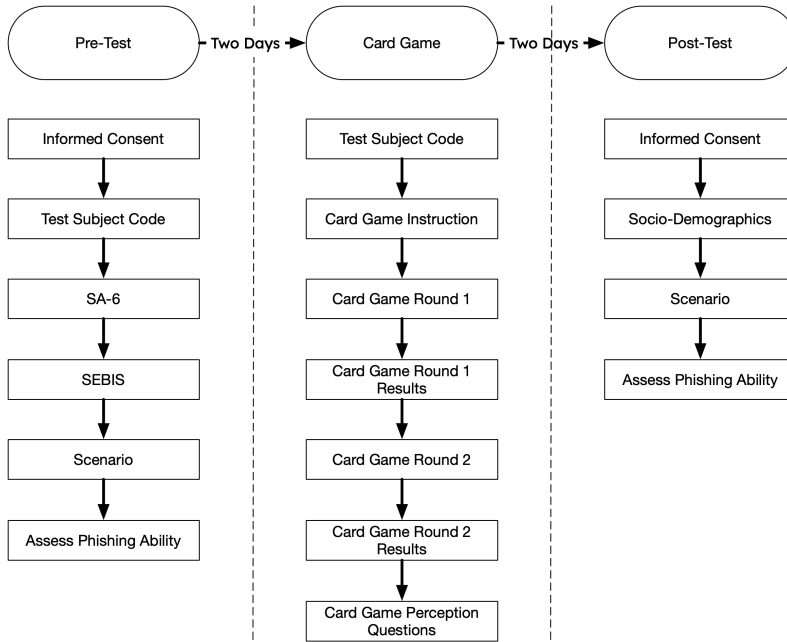


Fig. 4. Overview of pre-test, card game and post-test phase

but at least it allows for a certain significance, since only learning something once can lead to a lot of the learned content being already forgotten after the first night [56, 57].

3.2.5 Structure of Pre-Test and Post-Test. In this subsection, we want to present the structure of both the pre-test and post-test. In the following, we will explain the different steps in the flow of the online questionnaire. This includes the information in the heading indicating whether the respective step is only part of the pre-test or included in the pre-test and post-test. It also indicates how many pages of the online survey the step spans, e.g., just a single page or multiple pages. After the heading, each step will be explained in detail.

Step 1 - Informed Consent - pre-test only - page 1

First, participants were asked their consent to participate in the study (see Section 3.2.2 for details) and it was explained how data would be collected.

Step 2 - Test Subject Code - Both tests - page 2

Subsequently, the participants had to generate a test subject code. The purpose of this code was to ensure that anonymous attribution of the data from the pre-test, game phase and post-test was possible. Consequently, the participants provided a code with a procedure so that only they could know the code (see Section A for how the code was created) and the procedure should lead to the same result between all measurement time points.

Step 3 - Scales - pre-test only - page 3 and 4

The translated SA-6, and the SeBIS scales (see appendix B.3 for the items) were then asked. To the best of our knowledge and good faith, we translated the items. Each item was first translated into German by one person and then back-translated into English by a different person. In this process we made sure that both the wording and the meaning of the sentence remained as close as possible

to the original.

Step 4 - Scenario - Both tests - page 5

Before the test to assess the ability to distinguish between phishing and legitimate messages could begin, a short introductory scenario was given. Here, participants were told the number of examples presented and their task: “Your task is to examine each e-mail and determine whether if that is a legitimate e-mail or a phishing e-mail.” They were also asked to imagine a scenario, the purpose of which was to prevent them not to classify all providers as phishing based solely on personal experience or tendencies. The description of the scenario is as follows:

To complete the task, please assume the following scenario:

In order not to directly declare the senders, service providers and programs you have no connection to in reality as fraudulent, please assume in the following that:

You are Martin Müller and have the e-mail address: martin.mueller.77@gmail.com.

You use all services used in this questionnaire (YouTube, AirBnB, Apple, Amazon, Lufthansa, Google, LinkedIn, Microsoft, DHL, Netflix, GMX, PayPal, Vodafone, and HelloFresh).

Note: In this study, the links are disabled, so redirection is excluded.

Step 5 - Assess Phishing Ability - Both tests - page 6 to 26

Now participants saw the 20 examples (more on how the examples were created in Section 3.2.6) in a random order.

Step 6 - Socio Demographics - Post-test only - page 27

Finally, socio-demographic variables about gender, age, and formal education.

3.2.6 Tests Example Creation. The very first thing we did to create the examples was to look for a selection of well-known companies and their corresponding emails to customers. For this, we used companies from the list of the most visited domains in Germany. As far as possible, we have searched for emails that have been sent from these companies in exactly this form. These initially form the legitimate basis for the examples [58]. This eliminates or at least reduces the possibility that other factors than the ones we manipulated are used to assess whether the example is legitimate or phishing. The emails were all in German language, for demonstration purposes a translated version of the PayPal e-mail is depicted (see Figure 5). Similarly to other studies in the past, we decided to use the same number of phishing and legitimate examples. The actual distribution is unknown to the study participants. Therefore, they can never know whether the next example is one or the other. We looked for two examples per company and used one example as legitimate and one as phishing (see Table 1 for a listing of all examples).

The examples were converted to fraudulent messages. Either the sender address, the link URL, or the attachment format was changed to a fraudulent format. The changes were intended to provide as much coverage as possible of the content of the cards.

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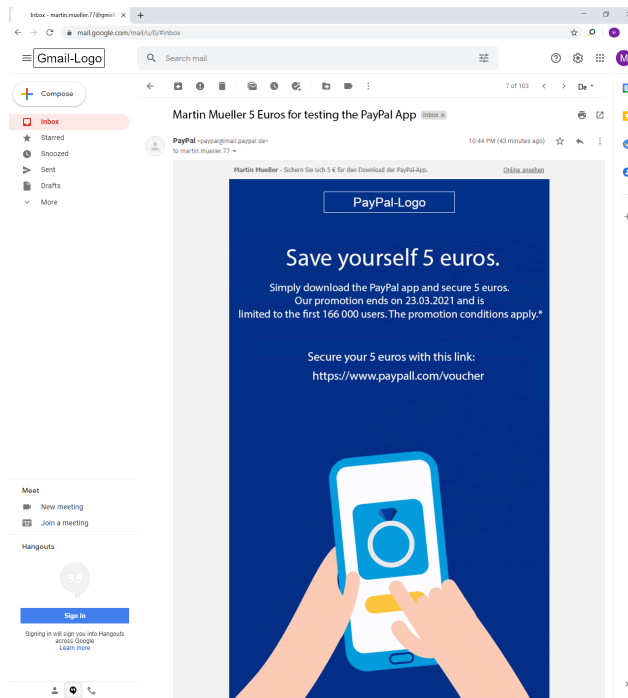


Fig. 5. Example of an e-mail used for assessing the ability to distinguish between phishing and legitimate messages. Note, we removed the logo before integrating it into the article.

Table 1. Overview of the Examples Matched to the Strategy Types, the Companies and the Card Game Cards

Company	Strategy Types	Sender / URL / Attachment Type	card game Match
YouTube	Legitimate	noreply-purchases@youtube.com	Cards 1 to 6
AirBnB	Legitimate	automated@airbnb.com	Cards 1 to 6
Apple	Legitimate	no_reply@email.apple.com	Cards 1 to 6
Amazon	Legitimate	https://packet.amazon.de/paketverfolgung	Cards 10,13,15,17,19,21,23
Lufthansa	Legitimate	https://www.lufthansa.com/buchungsanzeige	Cards 10,13,15,17,19,21,23
Google	Legitimate	https://www.google.com/neuesgerat	Cards 10,13,15,17,19,21,23
LinkedIn	Legitimate	https://video.linkedin.com/kurs	Cards 10,13,15,17,19,21,23
Microsoft	Legitimate	https://microsoft.com/speicher	Cards 10,13,15,17,19,21,23
DHL	Legitimate	https://mailing.dhl.de/wunschort...	Cards 10,13,15,17,19,21,23
Vodafone	Legitimate	PDF	Card 25
YouTube	Sender after @	noreply-purchases@youtube.com.uhszhiklo.cz	Cards 1 to 6
AirBnB	Random Sender	automated@host547.ru	Cards 1 to 6
Apple	Sender Typo	do_not_reply@email.aplle.com	Cards 1 to 6
Amazon	Random URL	https://telefon.host745.com/hinzufuegen	Card 10
Lufthansa	IP Adress	https://87.147.12.250/buchungsänderung	Card 10
Google	Deception before Domain	https://www.google.com.megahoust.ru/sicherheitscheck	Card 20
LinkedIn	Deception after Domain	https://login.linkyzt.com/www.linkedin.com/profil	Card 18
Microsoft	@ in Domain	https://microsoft.com@uhszhiklo.cz/default	Card 22
DHL	Typo	https://account.dlh.com/zustellung...	Card 24
HelloFresh	Malicious Attachment	EXE	Card 27

evaluating an individual's plans to perform specific security-related actions, whereas the SA-6 provides insight into the underlying cognitive and affective processes driving those actions. By considering both scales, researchers can gain a more comprehensive understanding of the complex relationships between human factors and security behaviors.

Table 2. Overview of Questions Used to Evaluate the Perception of the Card Game

Question	Answers
How did you like the game card game in terms of design?	Very bad Bad Rather bad Neither Rather good Good Very good
How comprehensible are the instructions of the game?	Very confusing Confusing Rather confusing Neither Rather understandable understandable very understandable
Due to the solution after the 1st round, I was more motivated in the 2nd round. To what extent do you agree with the preceding statement?	I do not agree at all I do not agree I rather disagree Neither I rather agree I agree I fully agree.
I feel better prepared to recognize possible phishing e-mails now that I have played the game. To what extent do you agree with the preceding statement?	I do not agree at all I do not agree I rather disagree Neither I rather agree I agree I fully agree.
I now feel confident in recognizing phishing emails. To what extent do you agree with the preceding statement?	I do not agree at all I do not agree I rather disagree Neither I rather agree I agree I fully agree.
A short assessment: How much new did you learn from the game?	Very little Little Rather little Neither Rather a lot A lot Very much.
What did you like about it?	Open Answer
What still needs to be improved?	Open Answer
Would you recommend the game to someone else?	Yes No

The inclusion of SeBIS and SA-6 in our study provides a more nuanced understanding of the human factors that influence security behaviors. By assessing both behavioral intentions (SeBIS) and attitudes towards security (SA-6), we can identify potential discrepancies between what individuals intend to do and their underlying attitudes, which may influence their actual behavior. For instance, an individual may indicate a high intention to use strong passwords (SeBIS), but hold a negative attitude towards security policies (SA-6), potentially leading to inconsistent behavior. This dual perspective allows us to better understand the complex psychological processes driving human behavior in the context of security and, ultimately, informs the design of more effective security training and awareness programs, such as our proposed gamified approach.

3.2.7 Structure of the Card Game Phase. The card game phase proceeded as follows. First, the course of the lab phase was explained to the participants, and possible questions were answered. A pre-developed text was used so all participants received the same information at first. The analog and digital phases were then alternated. First, participants again created their subject code on a study laptop. Followed by the game instructions and opportunity to read them in detail. After the participants had read the instructions and there were no further questions, the first initial round of the game started. Participants were given all the cards for round 1 and twelve minutes for completion. Meanwhile, the experimenter took notes on any noticeable behavior e. g. how the cards initially sorted by the participant (first sorting them all out first or starting directly with the top card). After round 1 was completed, the experimenter first took a picture of the solution. Then the participant could turn the cards over and see if their solution was correct or not. For wrong cases, they had the option to check which solution was correct instead. As next step, round 2 cards were dealt, and the participant started with the new cards. At the end of round 2, the experimenter took another picture of the result before the participant could check and reflect on their own solution. Subsequently, the experimenter uploaded the results to the online questionnaire in sosci survey and returned the laptop to the participant. The participant was then asked some explicit questions about their perception of the card game (see Table 2).

3.3 Data Analysis Approach

Phishing detection rate is an important measure when evaluating the effectiveness of interventions. But only looking at the phishing detection rate carries the risk of overlooking when study participants become overcautious due to a measure. For this reason, Signal Detection Theory [59] has already been used in various publications [60–71] to measure effectiveness. It provides two measures that are helpful for measuring effectiveness: (1) the sensitivity (d'), i. e. the ability to distinguish between legitimate and phishing messages, and (2) the criterion (C), i. e. the basic tendency towards legitimate or phishing messages. The sensitivity ranges from -3.38 to +3.38 (100%

Table 3. Statistical Summary of the Descriptive Values

Value	Min	Max	Median	Mean	SD
Overall Correct Answers pre-test	30	100	80	77.4	16.7
Phishing Correct Answers pre-test	30	100	90	83.1	17.8
Legitimate Correct Answers pre-test	60	100	90	88.2	13.9
Overall Correct Answers post-test	50	100	100	91.3	12.7
Phishing Correct Answers post-test	45	100	85	80.2	13.8
Legitimate Correct Answers post-test	55	100	90	89.8	10.1

correct answers). In contrast, a value of 0 represents the optimum for the criterion, and a positive (more legitimate) or negative value (more phishing) describes the tendency.

In addressing each research question, specific statistical tests were selected based on the nature of the data and the experimental design. Research Questions 1 and 3 utilize repeated measures ANOVA to examine within-subjects differences in sensitivity/criterion and certainty, respectively. A one-way ANOVA is used for Research Question 2 to assess between-group variations. For Research Question 4, Welch's one-way test was chosen due to its robustness in handling the heterogeneity of variances. In all cases, preliminary tests were conducted to verify that the data met the necessary assumptions for each analysis.

4 Results

The evaluation of "Hook & Match" aimed to assess its effectiveness in improving phishing detection skills while not adding more digital fatigue with an analog measure. To that end, we analyzed participants' ability to distinguish between phishing and legitimate examples before and after playing the game. In addition, we examined whether previous knowledge of phishing influenced learning outcomes. The following sections detail the key findings related to phishing detection effectiveness, the impact of previous knowledge, and user engagement with the game. First of all, please refer to Table 3 for the descriptive data on important values.

4.1 Sample

Of all the data collected, we had two steps in which participant data were excluded from further analysis. Step 1: No possibility to match the used subject code between pre-test and post-test. Step 2: Missing information about cards for round 1 and 2 of the card game. For step 1 we excluded two participants and for step 2 we excluded three participants.

In the end, data of 62 participants (36 male and 26 female) were analyzed. Of these participants, two were between 18 and 19, 45 between 20 and 24, ten between 25 and 29, three between 30 and 34 and two between 55 and 59 years old. One participant had no school education, 48 had finished high school and 13 had a university degree.

4.2 Improvement in Sensitivity Through the Card Game (RQ1)

We measure a sensitivity of 1.77 for the pre-test and a sensitivity of 2.48 for the post-test. Repeated measures ANOVA showed a statistically significant difference in the sensitivity between before playing the card game and after playing the card game ($F(1, 61) = 30.884, p < 0.001$) with an generalized eta-squared of $\eta_G^2 = 0.142$ (large effect).

There was not a statistically significant difference in the criterion between before playing the card game and after playing the card game ($F(1, 61) = 1.452, p = 0.233$) based on the repeated measures ANOVA.

Table 4. Improved Sensitivity for the Four Quartiles of Their Pre-Test Sensitivity

Group Comparison		Improved sensitivity		Significance
Group 1	Group 2	Mean Group 1	Mean Group 2	
1.Quartile	2.Quartile	1.87	0.593	**
1.Quartile	3.Quartile	1.87	0.258	***
1.Quartile	4.Quartile	1.87	0.23	***
2.Quartile	3.Quartile	0.593	0.258	ns
2.Quartile	4.Quartile	0.593	0.23	ns
3.Quartile	4.Quartile	0.258	0.23	ns

Quartile 1 would be the participants with the lowest pre-test sensitivity and quartile 4 those with the highest pre-test sensitivity. Also showing the significance level, “ns” for not significant, “****” for $p < 0.001$ and “***” for $p < 0.01$ and “**” for $p < 0.05$.

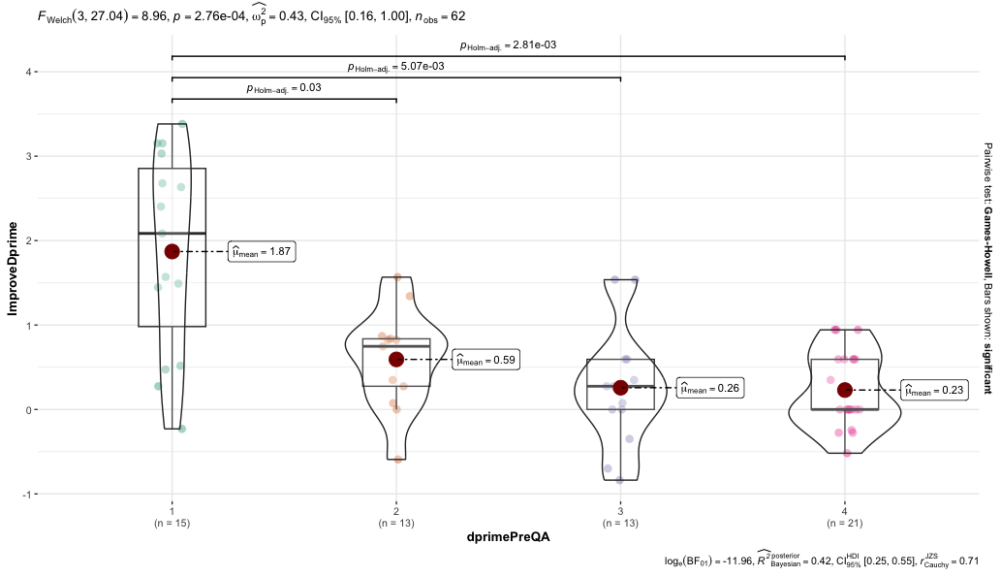


Fig. 6. Plot of the improved sensitivity for all four quartiles with the statistical values such as p -value and effect size

4.3 Impact of Initial Sensitivity on Card Game Effectiveness (RQ2)

We divided the participants into four quartiles with respect to their achieved scores for their pre-test sensitivity (see Table 4). Subsequently, these four quartiles were tested for differences as a group variable using a one-way ANOVA.

As the assumption of variance homogeneity was violated, we conducted a Welch one-way test and saw a significant difference ($p < 0.001$), $F(3,27) = 8.96$, $\omega_p^2 = 0.43$ (large effect). The post hoc group comparison Games-Howell correction can be found in Table 4. Also visible in Figure 6 the first quartile shows the highest improved sensitivity and is significantly higher than all the other three quartiles.

Table 5. Linear Regression for Improved Sensitivity Based on SA, SeBIS, Pre-Test Sensitivity and Card Game Results

Factor	Significance	Coefficient
(Intercept)	ns	-0.10456
SA6	ns	-0.02522
SeBIS	ns	0.00332
Sensitivity pre-test	***	-0.85935
Card game Result	***	0.11404

Also showing the significance level, “ns” for not significant, “****” for $p < 0.001$ and “***” for $p < 0.01$ and “*” for $p < 0.05$.

4.4 Effect of the Card Game on Confidence in Phishing Recognition (RQ3)

A repeated measures ANOVA was performed to compare the effect of playing the game on the certainty. There was a statistically significant difference in the certainty between before playing the card game and after playing the card game ($F(1, 61) = 19.28$, $p < 0.001$) with an generalized eta-squared of $\eta_G^2 = 0.088$ (medium effect). Participants have a certainty of 5.18 (SD = 1.28) prior to playing the card game and 5.89 (SD = 1.02) after playing on a scale from (1 = uncertain to 7 = very certain).

4.5 Influence of SA-6 and SeBIS Scores on Sensitivity Improvement (RQ4)

For RQ4, p -values for the predictor variables “Sensitivity pre-test” and “card game Result” are highly significant (see Table 5), so we can reject the null hypothesis and accept the alternative hypothesis, which means that there is a significant association between these predictors and the outcome variable improved sensitivity.

To further understand why both the SeBIS and SA-6 scales did not have a significant impact on improving sensitivity, the following approach was decided upon. We divided the participants into four quartiles with respect to their achieved scores for each of the SeBIS and the SA-6 scales (see Table 6). Subsequently, these four quartiles were tested for differences as a group variable using a one-way ANOVA.

First of all, we start with the SeBIS scale (possible values ranging from 16 to 80). The mean values for the four quartiles are as followed: Q1 = 42.6; Q2 = 52; Q3 = 59.4; Q4 = 67.6.

As the assumption of variance homogeneity was violated, we conducted a Welch one-way test and see no significant difference ($p = 0.222$), $F(3,29.3) = 1.55$, $\omega_p^2 = 0.05$ (small effect).

First of all, we start with the SA-6 scale (possible values ranging from 6 to 30). The mean values for the four quartiles are as followed: Q1 = 11.9; Q2 = 16.3; Q3 = 19.5; Q4 = 24.6.

As the assumption of variance homogeneity was violated, we conducted a Welch one-way test and see significant difference ($p = 0.016$), $F(3,30.82) = 4.02$, $\omega_p^2 = 0.21$ (large effect). The post hoc group comparison with Games-Howell correction can be found in Table 6.

The p -values for the predictor variables “SeBIS” and “card game result” are significant as shown in Table 6, so we can reject the null hypothesis and accept the alternative hypothesis, which means that there is a significant association between these predictors and the outcome variable Sensitivity pre-test (cf. Table 7).

Table 6. Improved Sensitivity for the Four Quartiles of their SA-6 Values

Group Comparison		Improved sensitivity		Significance
Group 1	Group 2	Mean Group 1	Mean Group 2	
1.Quartile	2.Quartile	1.5	0.289	*
1.Quartile	3.Quartile	1.5	0.766	ns
1.Quartile	4.Quartile	1.5	0.354	*
2.Quartile	3.Quartile	0.289	0.766	ns
2.Quartile	4.Quartile	0.289	0.354	ns
3.Quartile	4.Quartile	0.766	0.354	ns

Quartile 1 would be the participants with the lowest SA-6 value and quartile 4 those with the highest SA-6 value. Also showing the significance level, “ns” for not significant, “****” for $p < 0.001$ and “***” for $p < 0.01$ and “*” for $p < 0.05$.

Table 7. Linear Regression for Pre-Test Sensitivity Based on SA-6, SeBIS and Card Game Results

Factor	Significance	Coefficient
(Intercept)	***	-2.79748
SA6	ns	0.03331
SeBIS	*	0.02798
Card game Result	***	0.10413

4.6 Further Findings

Card matching. To assess the players’ matching, we visualized the questions and which answers the players chose as alluviation chart (see Figure 7). Numbers of the answers refer to the answer we had in mind when we designed the matching. Cards of the first round are depicted in a continuous coloring. Most cards show a low error rate. However, in particular card 03 (Sender 3) was switched with card 06 (Sender 6) and card 05 (Sender 5); and card 02 (Sender 2) was switched with card 13 (Who-Area 1). The cards of the second round are colored in pairs as these pairs reflected legitimate examples and a related example with a phishing url. This is also reflected in the errors, which were again low for most cards. However, players confused cards 23 (URL 5a) and 24 (URL 5b) and, to a lesser extent, cards 21 (URL 4a) and 22 (URL 4b).

Influence of perception on improved sensitivity. We wanted to compare for the perception questions if those people were different for their improvement of the phishing sensitivity. As shown in Figures 8, 9, 10 and 11 most answers are very positive. Even combining the answers into three categories like “positive”, “neutral” and “negative”, often only a handful of answers fall into the “negative” or “neutral” categories. Therefore, we could not conduct hypothesis testing for these research questions. We present the descriptive results. For all five questions the “positive” range from 73% (Motivation) to even 92% (Liked).

5 Discussion

Comparing our results for the sensitivity (pre-test $d' = 1.77$ and post-test $d' = 2.48$) with those of previous research we can see that our card game performs as well or better.

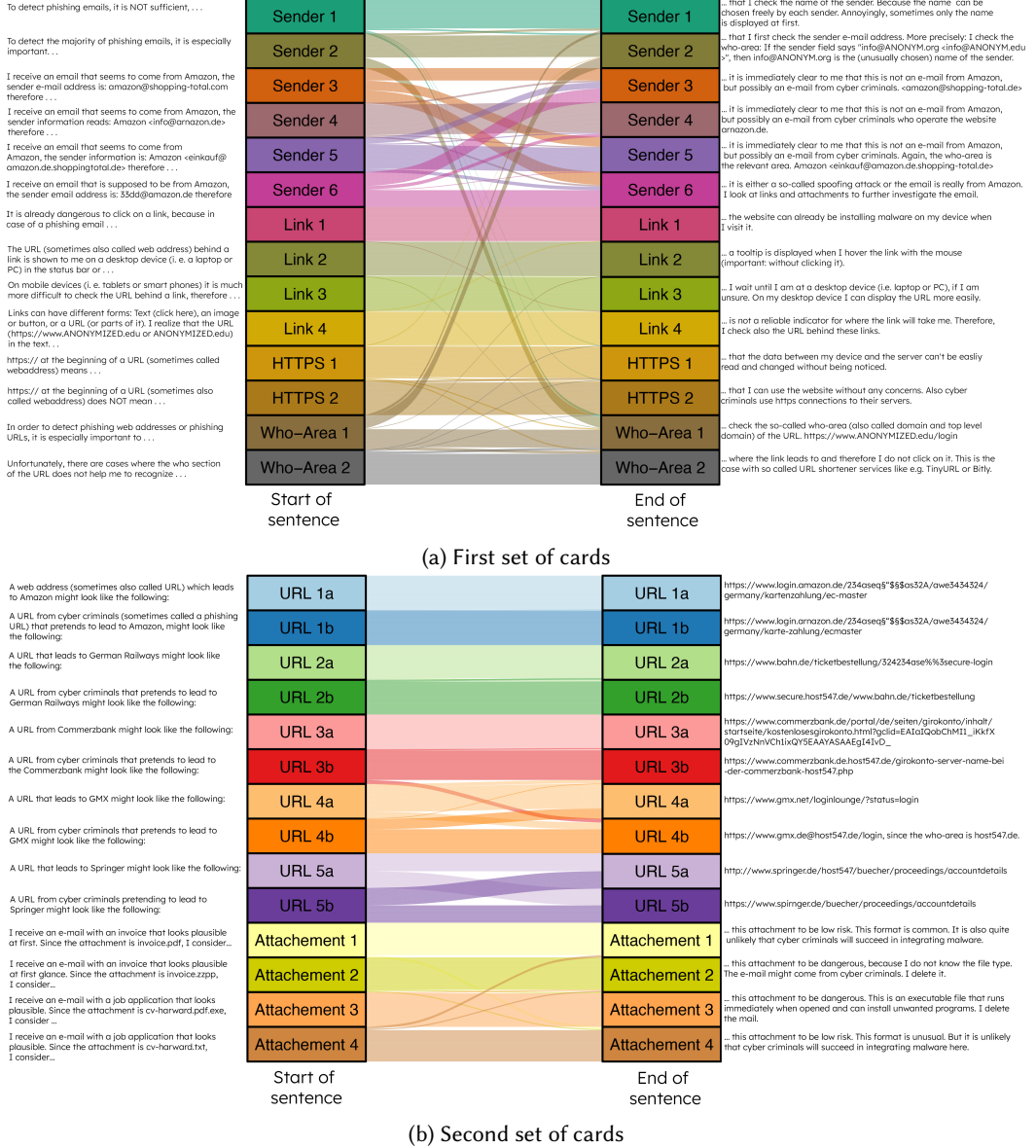


Fig. 7. Participants' results grouped by card in an alluviation chart.

Even though our post-test is two days after the intervention others like Reinheimer et al. [72] report $d' = 2.13$ for an instructor-based training. Sheng et al. [60] report a $d' = 2.02$ for their phishing game and another study from Berens et al. [73] report $d' = 2.12$ for their video. Only a more comprehensive online course from Berens et al. [74] with around three hours of estimated completion time reports slightly above the card game with an $d' = 2.66$. So, the card game performs equally or maybe even better than some way more extensive measure relatively immediately after playing. As discussed before it would be interesting to see if the active learning of the participants

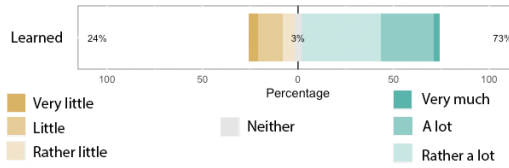


Fig. 8. Stacked bar chart for “A short assessment: How much new did you learn from the game?”

with the card game could improve knowledge retention and awareness even over a longer period. Especially since previous research found that people might recall such active learned knowledge better [75].

Looking at the Figure 7, one immediately notice that some cards have been assigned incorrectly. Here one must distinguish between the maps for round 1 and round 2. For round 1, there are several more complex interdependencies. For example, card 3 (“I receive an email that seems to come from Amazon, the sender e-mail address is: amazon@shop-ping-total.com therefore...”) is assigned to its matching counterpart (... it is immediately clear to me that this is not an e-mail from Amazon, but possibly an e-mail from cyber criminals.) as well as to the counterparts of cards 5 (“... it is immediately clear to me that this is not an e-mail from Amazon, but possibly an e-mail from cyber criminals. Here, too, the who area is the relevant area. Amazon <einkauf@amazon.de.shopping-total.de>”) and 6 (“... it is either a so-called spoofing attack or the email really comes from Amazon. I look at links and attachments to further investigate the e-mail”). So people apparently the missing indicator such as <amazon@shopping-total.de> for card 3 lead people to mistakenly use other cards.

Cards 2 (“To detect the majority of phishing emails, it is especially important...” and “... that I first check the sender e-mail address. More precisely: I check the who- area: If the sender field says “info@ANONYMIZED.ANONYMIZED.

ANONYMIZED.edu”, then info@ANONYMIZED.org is the (unusually chosen) name of the sender”) and 13 (“In order to detect phishing web addresses or phishing URLs, it is especially important to ...” and “... that I check the so-called who-area (also called domain and top level domain) of the URL. https://www.ANONYMIZED.ANONYMIZED.edu/login”) are also interchanged by some of the participants. Apparently the “especially important” was not strong enough and therefore lead to people not focusing on the URL, but instead also on the sender. Therefore, the improved card 13 would sound like “To identify phishing web addresses or URLs, the only reliable way is...”.

In the case of round 2, it is often the case that the legitimate and phishing examples by the same provider are swapped, e. g., cards 21 and 22 or 23 and 24.

The question now arises whether the cards should be adjusted for the future. We would argue that not all of these “misalignments” should be corrected for a future version—for several reasons: (1) If the solution is immediately obvious to each participant, there is a risk that the content will not be adequately reflected. For this to happen, there have to be difficult examples and it requires that mistakes are made. (2) Especially the examples from round 2, which are obviously phishing and legitimate, should not cause any ambiguity. In the future, it should be checked if the length of the game or the number of cards lead to carelessness, which could have an effect on these cards. On the other hand, we think that especially card 13 should be adjusted. After reflecting on the results, we have to conclude that we should be more concrete and absolute in our statements. This would be more in line with the basic tenor of the game and would probably allow for a simpler and clearer assignment. Accordingly, we suggest that the card should be changed to read “In order to detect phishing web addresses or phishing URLs, the only reliable indicator ... is the so-called who-area [...]”.

The question now arises why is it that only one of the two behavioral and attitudinal scales used appears to have a significant influence on the sensitivity of the pre-test. There are several possible factors that can be mentioned here. First, both scales are psychometric instruments used to measure security-related behavior. While the SA-6 scale, as its name suggests, focusses on awareness of security, SeBIS focusses on intentions regarding specific security-related behaviors. In psychology, awareness and intention are clearly distinguished from each other. Interestingly, one would initially assume that the SA-6 would be more appropriate for evaluation, since it is indeed a security awareness measure. A possible interpretation could be that, similar to psychology [76], awareness precedes intention in the decision-making process. And awareness represents an important step towards showing behavior, but only when intention is manifested can there be sufficient influence on sensitivity.

When looking at the results of other studies, similar results are shown, especially regarding the mean and standard deviation. For example, values of 54.62 [77] and 53.83 [78] were found in the past for the SeBIS scale. With a mean value of 56.19, the measured values move within almost the same range. The same applies to the SA-6, where values of 20.7 [79] and 21.53 [80] were measured in the past. Here too, the values measured in this study are at a very similar level with an average of 18.34. Accordingly, it seems that these values represent a widespread trend for the normal population. This reinforces once again that the statements made based on the results for SA-6 and SeBIS are at least based on results that have also been found in other studies.

5.1 Limitations

We have identified several limitations of our work.

Demographics influence on results. Participants in our study were on average 20 to 30 years old and had received a above-average level of education. Consequently, it is reasonable to assume that they are more inclined toward technical topics. Thus, the conclusions regarding the effectiveness of the card game can only be reliably applied to this particular group. Nonetheless, we believe that the card game, as an analog intervention based on a familiar principle, could also provide a easy entry point into the subject for older audiences. Moreover, our findings show that even participants with little prior knowledge experienced a substantial improvement in their ability to distinguish phishing from legitimate messages. These results suggest that other target groups may also gain value from our card game. Future studies should confirm these outcomes in different populations, such as an older cohort, to demonstrate that the findings hold more general significance.

Selection bias when sampling participants. Even though we tried to access people with a wide range of security awareness, we failed to recruit participants scoring in the bottom third of both scales. However, we are not aware of a provision of a distribution of the SeBIS or SA-6 scores for a representative sample. Thus, it might be possible that in a representative sample hardly any participants would be assigned to the bottom third.

Controlled environment in the laboratory. As we did the experiments in a laboratory, players were observed while playing the game. Of course, the players were aware of this, and may have acted differently than if they were playing the game at home or in their office. In particular, players might peek more and earlier on the backsides of the cards to find matching pairs when they do not feel observed. However, playing the game is considered to be voluntary, thus even if players peek from time to time, they still deal with the content of the game, and thus can benefit.

Translation of survey items. Since there was not an official translation available for most constructs, we translated the survey items. Although, we did a thorough translation, this might have led to a semantic change of the items challenging previous work on the the validity of the different

constructs. The problem of translating surveys is not new. Ervin and Bower [81] state that while in theory methodological considerations should be the only ones, in practice other matters like costs of translation cannot be ignored. As of now, it is not fully investigated how, when, and among whom language influences survey responses, although there seems to be consent that the survey results depend also on the used language [82]. Although their study was not on security awareness, [83] show that language can implicitly influence attitudes with a large effect size ($d = .72$). This also implies that the use of the original items in English could also have introduced a bias in the responses, in addition to the sample restrictions if we had to exclude non-English speaking participants.

Response and Social-desirability bias. As many of the survey items are self-reported and not directly measured, there might be a tendency of participants to answer questions in a way that is viewed favorably by others. However, one of the reasons why the participants were asked to complete the pre- and post-questionnaires two days before and after playing the game was to avoid reporting of short term stimuli which could only be observed immediately after the game.

Replay-ability of the card game. At this point we also wish to emphasize that our game lacks significant replay-ability. If players have already played the game once or twice, either to correct earlier mistakes or to self-test, they will not gain additional learning through further play. This stems from the deliberate decision to keep the game uncomplicated enough to serve as an accessible entry point for many; it is thus only an entry point for the topic. Interested participants must then turn to other measures. Existing options include a wide range of evaluated measures, from short videos to extended e-learning modules [84], which are predominantly digital. Accordingly, such in-depth measures should also be made available in analog form in the future, enabling an interested audience to deepen their understanding of phishing detection in a non-digital setting and to test their knowledge.

However, there are a couple of use cases where this is not an issue, in particular when there is only a short time frame available or the game mainly serves as an entry point for further discussions, e. g. at fairs and exhibitions, security circle trainings, congresses, and events. Since Hook & Match can be played unsupervised, is easy to learn and play, and has low manufacturing costs, it could also be used as a give-away and the backside could be used to show the organization's logo (cf. Figure 1).

6 Conclusion and Future Work

An educational card game, "Hook & Match", has been developed to enhance understanding of phishing attacks in a non-digital format. The game can be played individually and requires minimal instructions, making it accessible to a wide range of users (Section 3.1.1). Players match sentence beginnings with corresponding endings, allowing them to reflect on their mistakes and learn from them (Section 3.1.1). The game was evaluated in a laboratory study for its effectiveness in increasing sensitivity for phishing and for the perception of the game. It was shown that "Hook & Match" significantly improves the sensitivity for phishing and that this effect lasts for at least two days. The question for future research is how long does the effect of the card game last and whether there are differences in this retention time compared with other more detailed or passive forms of learning. It could also be measured if repeated play in different forms, such as after a certain amount of time or every few weeks, helps maintain the increased sensitivity [51]. The card game has also shown promising results, suggesting that it may be as effective or even more effective than other approaches (instructor-based trainings, games or online courses). For the future, it would be interesting to test different approaches against each other in an overall study and to focus even more on aspects of "game fun" and game duration, e. g. there are more complex and therefore longer lasting games [31]. In addition, valuable insights could be gained by investigating the impact

of prior knowledge from other security domains on improving sensitivity to phishing attacks. A larger and more diverse sample size may also reveal new findings related to the scales for SA-6 and SeBIS. While “Hook & Match” is an analog game by design, it can be easily adapted to digital contexts where needed or desired.

References

- [1] APWG. 2022. Phishing Activity Trends Report 4th Quarter 2022. (2022). Retrieved from [howpublished=https://docs.apwg.org/reports/apwg_trends_report_q4_2022.pdf](https://docs.apwg.org/reports/apwg_trends_report_q4_2022.pdf)
- [2] Paul Denny, Fiona McDonald, Ruth Empson, Philip Kelly, and Andrew Petersen. 2018. Empirical support for a causal relationship between gamification and learning outcomes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13. DOI : <http://dx.doi.org/10.1145/3173574.3173885>
- [3] Amon Rapp. 2018. Gamification for self-tracking. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. *Conference on Human Factors in Computing Systems* (2018). DOI : <http://dx.doi.org/10.1145/3173574.3173654>
- [4] Sam Scholefield and Lynsay A. Shepherd. 2019. Gamification techniques for raising cyber security awareness. *HCI for Cybersecurity, Privacy and Trust* (2019). DOI : <http://dx.doi.org/10.48550/arxiv.1903.08454>
- [5] Julia Lanzl, Tobias Manner-Romberg, Niclas Nüske, and Henner Gimpel. 2022. Digitaler stress in deutschland: Eine befragung von erwerbstätigen zu belastung und beanspruchung durch arbeit mit digitalen technologien. *Digitalisierung und Gesundheit* (2022), 265–302. DOI : <http://dx.doi.org/10.5771/9783748922933-265>
- [6] Michael Prince. 2004. Does active learning work? A review of the research. *Journal of Engineering Education* 93, 3 (2004), 223–231.
- [7] Fridolin S. T. Ting, Ronnie H. Shroff, Wai Hung Lam, Raycelle C. C. Garcia, Chi Lok Chan, Wing Ki Tsang, and Ndudi O. Ezeamuzie. 2023. A meta-analysis of studies on the effects of active learning on asian students’ performance in science, technology, engineering and mathematics (STEM) subjects. *The Asia-Pacific Education Researcher* 32, 3 (2023), 379–400. DOI : <http://dx.doi.org/10.1007/s40299-022-00661-6>
- [8] Ross H. Nehm, Stephen J. Finch, and Gena C. Sbeglia. 2022. Is active learning enough? the contributions of misconception-focused instruction and active-learning dosage on student learning of evolution. *BioScience* 72, 11 (2022), 1105–1117. DOI : <http://dx.doi.org/10.1093/biosci/biac073>
- [9] Mauricio Ronny De Almeida Souza, Lucas Furtini Veadó, Renata Teles Moreira, Eduardo Magno Lages Figueiredo, and Heitor Augustus Xavier Costa. 2017. Games for learning: Bridging game-related education methods to software engineering knowledge areas. In *2017 IEEE/ACM 39th International Conference on Software Engineering: Software Engineering Education and Training Track (ICSE-SEET)*. 170–179. DOI : <http://dx.doi.org/10.1109/ICSE-SEET.2017.17>
- [10] Peter Schaab, Kristian Beckers, and Sebastian Pape. 2016. A systematic Gap Analysis of Social Engineering Defence Mechanisms considering Social Psychology. In *10th International Symposium on Human Aspects of Information Security & Assurance, HAISA 2016, Frankfurt, Germany, July 19-21, 2016, Proceedings*. Retrieved from <https://www.cscan.org/openaccess/?paperid=301>
- [11] Melanie Volkamer, Karen Renaud, Benjamin Reinheimer, Philipp Rack, Marco Ghiglieri, Peter Mayer, Alexandra Kunz, and Nina Gerber. 2018. Developing and evaluating a five minute phishing awareness video. *Trust, Privacy and Security in Digital Business (TrustBus)* (2018), 119–134. DOI : http://dx.doi.org/10.1007/978-3-319-98385-1_9
- [12] Mauricio R.A. Souza, Lucas Veadó, Eduardo Figueiredo, Renata Teles Moreira, and Heitor Costa. 2017. Games for learning. *2017 IEEE/ACM 39th International Conference on Software Engineering: Software Engineering Education and Training Track (ICSE-SEET)* (2017), 170–179. DOI : <http://dx.doi.org/10.1109/icse-seet.2017.17>
- [13] Steve Sheng, Bryant Magnien, Ponnurangam Kumaraguru, Alessandro Acquisti, Lorrie Cranor, Jason Hong, and Elizabeth Nunge. 2007. Anti-phishing phil: The design and evaluation of a game that teaches people not to fall for phish. *Symposium on Usable Privacy and Security (SOUPS)* (2007).
- [14] Anirudh Ganesh, Chinenye Ndulue, and Rita Orji. 2023. Tailoring a persuasive game to promote secure smartphone behaviour. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (2023), 1–18. DOI : <http://dx.doi.org/10.1145/3544548.3581038>
- [15] Anirudh Ganesh, Chinenye Ndulue, and Rita Orji. 2021. Permarun- A persuasive game to improve user awareness and self-efficacy towards secure smartphone behaviour. *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (2021), 1–7. DOI : <http://dx.doi.org/10.1145/3411763.3451781>
- [16] Zikai Wen, Zhiqiu Lin, Rowena Chen, and Erik Andersen. 2019. What.Hack: Engaging anti-phishing training through a role-playing phishing simulation game. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (Conference on Human Factors in Computing Systems (CHI))*. 108. DOI : <http://dx.doi.org/10.1145/3290605.3300338>

- [17] Sebastian Pape, Ludger Goeke, Alejandro Quintanar, and Kristian Beckers. 2020. Conceptualization of a cybersecurity awareness quiz. In *Computer Security - ESORICS 2020 International Workshops MSTEC (LNCS)*, Vol. 12512. Springer International Publishing, Cham, 61–76. DOI : http://dx.doi.org/10.1007/978-3-030-62433-0_4
- [18] Dina Aladawy, Kristian Beckers, and Sebastian Pape. 2018. PERSUADED: Fighting social engineering attacks with a serious game. In *Trust, Privacy and Security in Digital Business - 15th International Conference, TrustBus 2018, Regensburg, Germany, September 5-6, 2018, Proceedings (Lecture Notes in Computer Science)*, Steven Furnell, Haralambos Mouratidis, and Günther Pernul (Eds.), Vol. 11033. Springer, 103–118. DOI : http://dx.doi.org/10.1007/978-3-319-98385-1_8
- [19] Gamze Canova, Melanie Volkamer, Clemens Bergmann, and Benjamin Reinheimer. 2015. NoPhish app evaluation: Lab and retention study. *USEC* (2015). DOI : <http://dx.doi.org/10.14722/usec.2015.23009>
- [20] Dorota Filipczuk, Charles Mason, and Stephen Snow. 2019. Using a game to explore notions of responsibility for cyber security in organisations. *Conference on Human Factors in Computing Systems* (2019), 1–6. DOI : <http://dx.doi.org/10.1145/3290607.3312846>
- [21] Ludger Goeke, Alejandro Quintanar, Kristian Beckers, and Sebastian Pape. 2019. PROTECT - An easy configurable serious game to train employees against social engineering attacks. In *Computer Security - ESORICS 2019 International Workshops, ISEC, MSTEC, and FINSEC, Luxembourg City, Luxembourg, September 26-27, 2019, Revised Selected Papers (Lecture Notes in Computer Science)*. Apostolos P. Fournaris, Manos Athanatos, Konstantinos Lampropoulos, Sotiris Ioannidis, George Hatzivasilis, Ernesto Damiani, Habtamu Abie, Silvio Ranise, Luca Verderame, Alberto Siena, and Joaquin Garcia-Alfaro (Eds.), Vol. 11981. Springer International Publishing, Cham, 156–171. DOI : http://dx.doi.org/10.1007/978-3-030-42051-2_11
- [22] Tamara Denning, Adam Lerner, Adam Shostack, and Tadayoshi Kohno. 2013. Control-alt-hack: The design and evaluation of a card game for computer security awareness and education. In *Proceedings of the 2013 ACM SIGSAC Conference on Computer & Communications Security*. 915–928.
- [23] Sylvain Frey, Awais Rashid, Pauline Anthonysamy, Maria Pinto-Albuquerque, and Syed Asad Naqvi. 2017. The good, the bad and the ugly: A study of security decisions in a cyber-physical systems game. *IEEE Transactions on Software Engineering* 45, 5 (2017), 521–536.
- [24] Kristian Beckers, Sebastian Pape, and Veronika Fries. 2016. HATCH: Hack and trick capricious humans – A serious game on social engineering. In *Proceedings of the 2016 British HCI Conference, Bournemouth, United Kingdom, July 11-15, 2016*. Retrieved from <https://ewic.bcs.org/content/ConWebDoc/56973>
- [25] Kristian Beckers and Sebastian Pape. 2016. A serious game for eliciting social engineering security requirements. In *Proceedings of the 24th IEEE International Conference on Requirements Engineering (RE '16)*. IEEE Computer Society, 16–25. DOI : <http://dx.doi.org/10.1109/RE.2016.39>
- [26] Andreas Rieb and Ulrike Lechner. 2016. Operation digital chameleon: Towards an open cybersecurity method. In *Proceedings of the 12th International Symposium on Open Collaboration*. 1–10.
- [27] Stephen Hart, Andrea Margheri, Federica Paci, and Vladimiro Sassone. 2020. Riskio: A serious game for cyber security awareness and education. *Computers & Security* 95 (2020), 101827.
- [28] Emergent Network Defense. Emergyn Risk Homepage. Retrieved from <https://emergynt.com/risk-deck/>. ([n. d.]).
- [29] Carsten Schulte, Brett A. Becker, Monica Divitini, Erik Barendsen, Maria Hwang, and Mark Santolucito. 2021. cardComposer: A functional programming card game. *Proceedings of the 26th ACM Conference on Innovation and Technology in Computer Science Education V. 2* (2021), 635–636. DOI : <http://dx.doi.org/10.1145/3456565.3460029>
- [30] Rubia Fatima, Affan Yasin Chouhan, Lin Liu, and Jianmin Wang. 2022. What should abeeha do? an activity for phishing awareness. In *22nd International Conference on Software Quality, Reliability, and Security (QRS)*.
- [31] Rubia Fatima, Affan Yasin, Lin Liu, and Jianmin Wang. 2019. How persuasive is a phishing email? A phishing game for phishing awareness. *Journal of Computer Security* 27, 6 (2019), 581–612.
- [32] Malcolm Knowles. 1977. Adult learning processes: Pedagogy and andragogy. *Religious Education* 72, 2 (1977), 202–211.
- [33] Albert Bandura. 1997. *Self-efficacy: The Exercise of Control*. Macmillan.
- [34] John Dewey. 1986. Experience and education. In *The Educational Forum*, Vol. 50. Taylor & Francis, 241–252.
- [35] Cheryl I Johnson and Richard E Mayer. 2009. A testing effect with multimedia learning. *Journal of Educational Psychology* 101, 3 (2009), 621.
- [36] John Sweller. 1988. Cognitive load during problem solving: Effects on learning. *Cognitive Science* 12, 2 (1988), 257–285.
- [37] Staffan Noteberg. 2009. Pomodoro technique illustrated: The easy way to do more in less time. *Pomodoro Technique Illustrated* 1 (2009), 1–156.
- [38] Federico Gobbo and Matteo Vaccari. 2008. The pomodoro technique for sustainable pace in extreme programming teams. In *International Conference on Agile Processes and Extreme Programming in Software Engineering*. Springer, 180–184.
- [39] Maria Bada, Angela M. Sasse, and Jason R. C. Nurse. 2019. Cyber security awareness campaigns: Why do they fail to change behaviour? *CoRR abs/1901.02672* (2019). *arXiv preprint arXiv:1901.02672*. Retrieved from <https://arxiv.org/abs/1901.02672>

- [40] Janet M. Carrington. 1978. The games biochemist play. *Biochemical Education* 6, 4 (1978), 80–81.
- [41] Gamze Canova, Melanie Volkamer, Clemens Bergmann, Roland Borza, Benjamin Reinheimer, Simon Stockhardt, and Ralf Tenberg. 2015. Learn to spot phishing urls with the android nophish app. In *WISE 9*. Springer, 87–100.
- [42] Mia Consalvo, Thorsten Busch, and Carolyn Jong. 2019. Playing a better me: How players rehearse their ethos via moral choices. *Games and Culture* 14, 3 (2019), 216–235.
- [43] Glena H. Iten, Sharon T. Steinemann, and Klaus Opwis. 2018. Choosing to help monsters: A mixed-method examination of meaningful choices in narrative-rich games and interactive narratives. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [44] Michael D. Slater and Donna Rouner. 2002. Entertainment—education and elaboration likelihood: Understanding the processing of narrative persuasion. *Communication Theory* 12, 2 (2002), 173–191.
- [45] Paschal Sheeran. 2002. Intention—behavior relations: A conceptual and empirical review. *European Review of Social Psychology* 12, 1 (2002), 1–36.
- [46] Albert Bandura and Nancy E Adams. 1977. Analysis of self-efficacy theory of behavioral change. *Cognitive Therapy and Research* 1, 4 (1977), 287–310.
- [47] Ka Chung Ng, Xiaojun Zhang, James Y.L. Thong, and Kar Yan Tam. 2021. Protecting against threats to information security: An attitudinal ambivalence perspective. *Journal of Management Information Systems* 38, 3 (2021), 732–764.
- [48] Serge Egelman and Eyal Peer. 2015. Scaling the security wall: Developing a security behavior intentions scale (sebis). In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 2873–2882.
- [49] Serge Egelman, Marian Harbach, and Eyal Peer. 2016. Behavior ever follows intention? A validation of the Security Behavior Intentions Scale (SeBIS). In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 5257–5261.
- [50] Cori Faklaris, Laura A. Dabbish, and Jason I. Hong. 2019. A {self-report} measure of {end-user} security attitudes (SA-6). In *Fifteenth Symposium on Usable Privacy and Security (SOUPS 2019)*. 61–77.
- [51] Sebastian Pape. 2021. Challenges for designing serious games on security and privacy awareness. In *Privacy and Identity Management. Between Data Protection and Security - 16th IFIP WG 9.2, 9.6/11.7, 11.6/SIG 9.2.2 International Summer School, Privacy and Identity 2021, Virtual Event, August 16-20, 2021, Revised Selected Papers*, Michael Friedewald, Stephan Krenn, Ina Schiering, and Stefan Schiffner (Eds.). IFIP Advances in Information and Communication Technology, Vol. 644. Springer, Chapter Challenges for Designing Serious Games on Security and Privacy Awareness, 3–16. DOI: http://dx.doi.org/10.1007/978-3-030-99100-5_1
- [52] Jaap M. J. Murre and Joeri Dros. 2015. Replication and analysis of ebbinghaus' forgetting curve. *PLoS ONE* 10, 7 (2015), e0120644. DOI: <http://dx.doi.org/10.1371/journal.pone.0120644>
- [53] Nelson Cowan. 2008. Chapter 20 what are the differences between long-term, short-term, and working memory? *Progress in Brain Research* 169 (2008), 323–338. DOI: [http://dx.doi.org/10.1016/S0079-6123\(07\)00020-9](http://dx.doi.org/10.1016/S0079-6123(07)00020-9)
- [54] Henry L. Roediger and Andrew C. Butler. 2011. The critical role of retrieval practice in long-term retention. *Trends in Cognitive Sciences* 15, 1 (2011), 20–27.
- [55] Benjamin M. Berens, Mattia Mossano, and Melanie Volkamer. 2023. Taking 5 minutes protects you for 5 months: Evaluating an anti-phishing awareness video. *Computers & Security* (2023), 103620. DOI: <http://dx.doi.org/10.1016/j.cose.2023.103620>
- [56] Jared M. Saletin, Andrea N. Goldstein, and Matthew P. Walker. 2011. The role of sleep in directed forgetting and remembering of human memories. *Cerebral Cortex* 21, 11 (2011), 2534–2541. DOI: <http://dx.doi.org/10.1093/cercor/bhr034>
- [57] Luisa de Vivo, Michele Bellesi, William Marshall, Eric A. Bushong, Mark H. Ellisman, Giulio Tononi, and Chiara Cirelli. 2017. Ultrastructural evidence for synaptic scaling across the wake/sleep cycle. *Science* 355, 6324 (2017), 507–510. DOI: <http://dx.doi.org/10.1126/science.aah5982>
- [58] Benjamin Maximilian Berens, Florian Schaub, Mattia Mossano, and Melanie Volkamer. 2024. Better together: The interplay between a phishing awareness video and a link-centric phishing support tool. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 826, 60 pages. DOI: <http://dx.doi.org/10.1145/3613904.3642843>
- [59] Harold Stanislaw and Natasha Todorov. 1999. Calculation of signal detection theory measures. *Behavior Research Methods, Instruments, & Computers* 1, 31 (1999), 137–149. Retrieved from <https://link.springer.com/content/pdf/10.3758/BF03207704.pdf>
- [60] Steve Sheng, Bryant Magnien, Ponnuram Kumaraguru, Alessandro Acquisti, Lorrie Faith Cranor, Jason Hong, and Elizabeth Nunge. 2007. Anti-phishing phil: The design and evaluation of a game that teaches people not to fall for phishing. In *Symposium on Usable Privacy and Security (SOUPS)*. ACM, 88–99.
- [61] Iain Embrey and Kim Kaivanto. 2023. Many phish in the C: A coexisting-choice-criteria model of security behavior. *Risk Analysis* 43, 4 (2023), 783–799.
- [62] Casey Inez Canfield and Baruch Fischhoff. 2018. Setting priorities in behavioral interventions: An application to reducing phishing risk. *Risk Analysis* 38 (2018), 826–838. DOI: <http://dx.doi.org/10.1111/risa.12917>

- [63] Jaclyn Martin, Chad Dubé, and Michael D. Coovert. 2018. Signal detection theory (sdt) is effective for modeling user behavior toward phishing and spear-phishing attacks. *Human Factors: The Journal of Human Factors and Ergonomics Society* 60 (2018), 1179–1191. DOI : <http://dx.doi.org/10.1177/0018720818789818>
- [64] Maria M. Moreno-Fernández, Fernando Blanco, Pablo Garaizar, and Helena Matute. 2017. Fishing for phishers. Improving internet users' sensitivity to visual deception cues to prevent electronic fraud. *Computers in Human Behavior* 69 (2017), 421–436. DOI : <http://dx.doi.org/10.1016/j.chb.2016.12.044>
- [65] Casey Canfield, Alex Davis, Baruch Fischhoff, and Forget A on Usable 2017. Replication: Challenges in using data logs to validate phishing detection ability metrics. *Symposium on Usable Privacy and Security (SOUPS)* (2017).
- [66] M. Butavicius, K. Parsons, M. Pattinson, A. McCormac, D. Calic, and M. Lillie. 2017. Understanding susceptibility to phishing emails: Assessing the impact of individual differences and culture. *International Symposium on Human Aspects of Information Security & Assurance (HAISA'17)*.
- [67] Marcus Butavicius, Kathryn Parsons, Malcolm Pattinson, and Agata McCormac. 2016. Breaching the human firewall: Social engineering in phishing and spear-phishing emails. *Australasian Conference on Information Systems* (2016).
- [68] Casey Canfield, Baruch Fischhoff, and Alex Davis. 2016. Quantifying phishing susceptibility for detection and behavior decisions. *Human Factors: The Journal of Human Factors and Ergonomics Society* 58 (2016), 1158–1172. DOI : <http://dx.doi.org/10.1177/0018720816665025>
- [69] Casey Canfield, Baruch Fischhoff, and Alex Davis. 2015. Using signal detection theory to measure phishing detection ability and behavior. In *SOUPS*.
- [70] Christopher B. Mayhorn and Patrick G. Nyeste. 2012. Training users to counteract phishing. *Work (Reading, Mass.)* 41 Suppl 1 (2012), 3549–52. DOI : <http://dx.doi.org/10.3233/wor-2012-1054-3549>
- [71] Jaclyn Martin. 2017. *Something Looks Phishy Here: Applications of Signal Detection Theory to Cyber-Security Behaviors in the Workplace*. Ph.D. Dissertation. University of South Florida.
- [72] Benjamin Reinheimer, Lukas Aldag, Peter Mayer, Mattia Mossano, Reyhan Duezguen, Bettina Lofthouse, Tatiana von Landesberger, and Melanie Volkamer. 2020. An investigation of phishing awareness and education over time: When and how to best remind users. In *Symposium on Usable Privacy and Security (SOUPS)*. 259–284.
- [73] Benjamin Maximilian Berens, Florian Schaub, Mattia Mossano, and Melanie Volkamer. 2024. Better together: The interplay between a phishing awareness video and a link-centric phishing support tool. *Proceedings of the CHI Conference on Human Factors in Computing Systems* (2024), 1–60. DOI : <http://dx.doi.org/10.1145/3613904.3642843>
- [74] Benjamin Berens, Mattia Mossano, and Melanie Volkamer. 2022. Phishing awareness and education - When to best remind?. In *Symposium on Usable Security and Privacy (USEC) (Symposium on Usable Security and Privacy (USEC))*, Vol. USEC. Internet Society, San Diego, CA, USA, 1–15.
- [75] Josh B. Kurtz, Michael A. Lourie, Elizabeth E. Holman, Karri L. Grob, and Seetha U. Monrad. 2019. Creating assessments as an active learning strategy: what are students' perceptions? A mixed methods study. *Medical Education Online* 24, 1 (2019), 1630239. DOI : <http://dx.doi.org/10.1080/10872981.2019.1630239>
- [76] Icek Ajzen. 1991. The theory of planned behavior. *Organizational Behavior and Human Decision Processes* (1991).
- [77] Christopher Conetta. 2019. Individual differences in cyber security. *McNair Research Journal SJSU* 15, 1 (2019), 4.
- [78] Daniel Kelley. 2018. Investigation of attitudes towards security behaviors. *McNair Research Journal SJSU* 14, 1 (2018), 10.
- [79] Desiree Abrokwa, Shruti Das, Omer Akgul, and Michelle L. Mazurek. 2021. Comparing security and privacy attitudes among {us}. Users of different smartphone and {smart-speaker} platforms. In *Seventeenth Symposium on Usable Privacy and Security (SOUPS 2021)*. 139–158.
- [80] Amani Mohammed Alqarni, Daniel Timko, and Muhammad Lutfor Rahman. 2023. Saudi arabian perspective of security, privacy, and attitude of using facial recognition technology. In *2023 20th Annual International Conference on Privacy, Security and Trust (PST)*. IEEE, 1–12.
- [81] Susan Ervin and Robert T. Bower. 1952. Translation problems in international surveys. In *Public Opinion Quarterly* 16, 4 (1952), 595–604. DOI : <http://dx.doi.org/10.1086/266421>
- [82] Efrén O. Pérez. 2015. The language-opinion connection. In *The Oxford Handbook of Polling and Survey Methods* (oxford uni ed.), Lonna Rae Atkeson and R. Michael Alvarez (Eds.). Oxford University Press. DOI : <http://dx.doi.org/10.1093/oxfordhb/9780190213299.013.18>
- [83] Oludamini Ogunning, Yarrow Dunham, and Mahzarin R. Banaji. 2010. The language of implicit preferences. *Journal of Experimental Social Psychology* 46, 6 (nov 2010), 999–1003. DOI : <http://dx.doi.org/10.1016/J.JESP.2010.07.006>
- [84] Simon Stockhardt, Benjamin Reinheimer, Melanie Volkamer, Peter Mayer, Alexandra Kunz, Philipp Rack, and Daniel Lehmann. 2016. Teaching phishing-security: Which way is best? *International Conference on ICT Systems Security and Privacy Protection (IFIP SEC)* (2016), 135–149. DOI : http://dx.doi.org/10.1007/978-3-319-33630-5_10

Appendix

A Test Subject Code Text

We used the following instructions:

Dear participants since we want to assign your questionnaire anonymously, it is important that you generate your personal code. This is the only way that your questionnaires can be assigned to each other without anyone being able to find out who filled out these questionnaires. So it is important that you follow the code generation process exactly. You do not have to remember the code, because the same instructions are provided to you again later on.

Please name the first letter of your father's first name (or a person who is closest to a father for you). (e. g., Anton, Bernd, Hans-Peter, etc.)

Please name the first letter of your mother's first name (or a person who is closest to a mother for you). (e. g., Anna, Beate, Jutta, Maria, etc.)

Please indicate the first letter of your first name (e. g., Michael, Thomas, Ute, etc.).

Please state the day of your birth date (e. g., birthday on January 7 = 7, on May 12 = 12, on October 31 = 31).

Please state the last letter of your natural hair color. (e. g., brown, bald, black, etc.).

Please state the last letter of your eye color. (e. g., brown, green, gray, etc.).

Please state the last letter of your last name (If you have changed your name, use your birth name!).

B Questionnaire Items (Original and Translated)

B.1 SeBIS

Security Behaviour Intention Scale with scale Never (1), Rarely (2), Sometimes (3), Often (4), and Always (5) [48, 49]

- I set my computer screen to automatically lock if I do not use it for a prolonged period of time.
- I use a password/passcode to unlock my laptop or tablet.
- I manually lock my computer screen when I step away from it.
- I use a PIN or passcode to unlock my mobile phone.
- I do not change my passwords, unless I have to.
- I use different passwords for different accounts that I have.
- When I create a new online account, I try to use a password that goes beyond the site's minimum requirements.
- I do not include special characters in my password if it's not required.
- When someone sends me a link, I open it without first verifying where it goes.
- I know what website I'm visiting based on its look and feel, rather than by looking at the URL bar.
- I submit information to websites without first verifying that it is sent securely (e. g. SSL, "https://", a lock icon).
- When browsing websites, I mouse-over links to see where they go, before clicking them.
- If I discover a security problem, I continue what I was doing because I assume someone else is going to fix it.
- When I'm prompted about a software update, I install it right away.
- I try to make sure that the programs I use are up-to-date.
- I verify that my anti-virus software has been regularly updating itself.

B.2 SA-6

A six-item scale for assessing people's security attitudes with Likert-type agreement scale (1=Strongly disagree, 5=Strongly agree) [50]

- I seek out opportunities to learn about security measures that are relevant to me.
- I am extremely motivated to take all the steps needed to keep my online data and accounts safe.
- Generally, I diligently follow a routine about security practices.
- I often am interested in articles about security threats.
- I always pay attention to experts' advice about the steps I need to take to keep my online data and accounts safe.
- I am extremely knowledgeable about all the steps needed to keep my online data and accounts safe.

B.3 SeBIS Translated

Security Behaviour Intention Scale with scale Niemals (1), Selten (2), Manchmal (3), Oft (4), and Immer (5)

- Ich stelle meinen Computerbildschirm so ein, dass er automatisch gesperrt wird, wenn ich ihn über einen längeren Zeitraum nicht benutze.
- Ich verwende ein Passwort/einen Passcode, um meinen Laptop oder mein Tablet zu entsperren.
- Ich sperre meinen Computerbildschirm manuell, wenn ich mich von ihm entferne.
- Ich benutze eine PIN oder einen Passcode, um mein Mobiltelefon zu entsperren.
- Ich ändere meine Passwörter nicht, es sei denn, ich muss es.
- Ich verwende verschiedene Passwörter für verschiedene Konten, die ich habe.
- Wenn ich ein neues Online-Konto einrichte, versuche ich, ein Passwort zu verwenden, das über die Mindestanforderungen der Website hinausgeht.
- Ich verwende keine Sonderzeichen in meinem Passwort, wenn dies nicht erforderlich ist.
- Wenn mir jemand einen Link schickt, öffne ich ihn, ohne vorher zu überprüfen, wohin er führt.
- Ich erkenne die Website, die ich besuche, anhand ihres Aussehens und ihrer Haptik und nicht durch einen Blick auf die URL-Leiste.
- Ich sende Informationen an Websites, ohne vorher zu überprüfen, ob sie sicher übertragen werden (z. B. SSL, „https://“, ein Schlosssymbol).
- Wenn ich auf Webseiten surfe, fahre ich mit der Maus über Links, um zu sehen, wohin sie führen, bevor ich sie anklicke.
- Wenn ich ein Sicherheitsproblem entdecke, setze ich meine Arbeit fort, weil ich davon ausgehe, dass jemand anderes es beheben wird.
- Wenn ich zu einem Software-Update aufgefordert werde, installiere ich es sofort.
- Ich versuche sicherzustellen, dass die Programme, die ich verwende, auf dem neuesten Stand sind.
- Ich überprüfe, ob meine Antiviren-Software regelmäßig aktualisiert wird.

B.4 SA-6 Translated

A six-item scale for assessing people's security attitudes with Likert-type agreement scale (1=stimme gar nicht zu, 5=stimme voll zu)

- Ich suche nach Möglichkeiten, mich über Sicherheitsmaßnahmen zu informieren, die für mich relevant sind.

- Ich bin äußerst motiviert, alle notwendigen Schritte zu unternehmen, um meine Online-Daten und -Konten zu schützen.
- Im Allgemeinen folge ich gewissenhaft einer Routine für Sicherheitspraktiken.
- Ich bin interessiert mich oft für Artikeln über Sicherheitsbedrohungen.
- Ich höre immer auf die Ratschläge von Experten, welche Schritte ich unternehmen muss, um meine Online-Daten und -Konten sicher zu halten.
- Ich bin sehr gut informiert über alle Schritte, die nötig sind, um meine Online-Daten und -Konten sicher zu halten.

C English Card Texts (Translated)

Table 8. Translation of the First Set's Cards' Texts

No.	Head of sentence	Tail of sentence	Category
1	To detect phishing emails, it is NOT sufficient, ...	...that I check the name of the sender. Because the name can be chosen freely by each sender. Annoyingly, sometimes only the name is displayed at first.	Sender 1
2	To detect the majority of phishing emails, it is especially important...	...that I first check the sender e-mail address. More precisely: I check the who- area: If the sender field says "info@ANONYMIZED.org" then info@ANONYMIZED.org is the (unusually chosen) name of the sender.	Sender 2
3	I receive an email that seems to come from Amazon, the sender e-mail address is: amazon@shop-ping-total.com therefore ...	...it is immediately clear to me that this is not an e-mail from Amazon, but possibly an e-mail from cyber criminals.	Sender 3
4	I receive an email that seems to come from Amazon, the sender information reads: Amazon <info@arnazon.de> therefore ...	...it is immediately clear to me that this is not an e-mail from Amazon, but possibly an e-mail from cyber criminals who operate the website arnazon.de website.	Sender 4
5	I receive an email that seems to come from Amazon, the sender information is: Amazon <einkauf@amazon.de.shoppingtotal.de> therefore ...	...it is immediately clear to me that this is not an e-mail from Amazon, but possibly an e-mail from cyber criminals. Again, the who-area is the relevant area. Amazon <einkauf@amazon.de.shopping-total.de>	Sender 5
6	I receive an email that is supposed to be from Amazon, the sender email address is: 33dd@amazon.de therefore ...	...it is either a so-called spoofing attack or the email is really from Amazon. I look at links and attachments to further investigate the email.	Sender 6
7	It is already dangerous to click on a link, because in case of a phishing email ...	...the website can already be installing malware on my device when I visit it.	Link 1
8	The URL (sometimes also called web address) behind a link is shown to me on a desktop device (i. e. a laptop or PC) in the status bar or ...	...a tooltip is displayed when I hover the link with the mouse (important: without clicking it).	Link 2
9	On mobile devices (i. e. tablets or smart phones) it is much more difficult to check the URL behind a link, therefore ...	...I wait until I am at a desktop device (i.e. laptop or PC), if I am unsure. On my desktop device I can display the URL more easily.	Link 3
10	Links can have different forms: Text (click here), an image or button, or a URL (or parts of it). I realize that the URL (https://www.ANONYMIZED.edu or ANONYMIZED.edu) in the text ...	...is not a reliable indicator for where the link will take me. Therefore, I check also the URL behind these links.	Link 4
11	https:// at the beginning of a URL (sometimes called webaddress) means ...	...that the data between my device and the server cannot be easily read and changed without being noticed.	HTTPS 1
12	https:// at the beginning of a URL (sometimes also called webaddress) does NOT mean ...	...that I can use the website without any concerns. Also cyber criminals use https connections to their servers.	HTTPS 2
13	In order to detect phishing web addresses or phishing URLs, it is especially important to ...	...that I check the so-called who-area (also called domain and top level domain) of the URL. https://www.ANONYMIZED.ANONYMIZED.edu/login	Who-Area 1
14	Unfortunately, there are cases where the who section of the URL does not help me to recognize ...	...where the link leads to and therefore I do not click on it. This is the case with so called URL shortener services like e.g. TinyURL or Bitly.	Who-Area 2

Table 9. Translation of the Second Set's Cards' Texts

No.	Head of sentence	Tail of sentence	Category
15	A web address (sometimes also called URL) which leads to Amazon might look like the following:	https://www.login.amazon.de/234aseq\$\$\$as32A/awe3434324/germany/kartenzahlung/ec-master	URL 1a
16	A URL from cyber criminals (sometimes called a phishing URL) that pretends to lead to Amazon, might look like the following:	https://www.login.arnazon.de/234aseq\$\$\$as32A/awe3434324/germany/karte-zah-lung/ecmaster	URL 1b
17	A URL that leads to German Railways might look like the following:	https://www.bahn.de/ticketbestellung/324234ase%3secure-login	URL 2a
18	A URL from cyber criminals that pretends to lead to German Railways might look like the following:	https://www.secure.host547.de/www.bahn.de/ticketbestellung	URL 2b
19	A URL from Commerzbank might look like the following:	https://www.commerzbank.de/portal/de/seiten/girokonto/inhalt/startseite/kostenlos Girokonto.html?gclid=EAlaQobChMI_iKkFX09glVzNnVCh1ixQY5EAAAYASAAEgI4lvD_BwE	URL 3a
20	A URL from cyber criminals that pretends to lead to the Commerzbank might look like the following:	https://www.commerzbank.de/host547.de/girokonto-server-name-bei-der-commerzbank-host547.php	URL 3b
21	A URL that leads to GMX might look like the following:	https://www.gmx.net/loginlounge/?status=login	URL 4a
22	A URL from cyber criminals that pretends to lead to GMX might look like the following:	https://www.gmx.de/host547.de/login/	URL 4b
23	A URL that leads to Springer might look like the following:	http://www.springer.de/host547/buecher/proceedings/accountdetails	URL 5a
24	A URL from cyber criminals pretending to lead to Springer might look like the following:	https://www.spirnger.de/buecher/proceedings/accountdetails	URL 5b
25	I receive an e-mail with an invoice that looks plausible at first. Since the attachment is invoice.pdf, I consider ...	...this attachment to low risk. This format is common. It is also quite unlikely that cyber criminals will succeed in integrating malware.	Attachment 1
26	I receive an e-mail with an invoice that looks plausible at first. Since the attachment is invoice.zzpp, I consider ...	...this attachment to be dangerous, because I do not know the file type. The e-mail might come from cyber criminals. I delete it.	Attachment 2
27	I receive an e-mail with a job application that looks plausible. Since the attachment is cv-harward.pdf.exe, I consider ...	...this attachment to be dangerous. This is an executable file that runs immediately when opened and can install unwanted programs. I delete the mail.	Attachment 3
28	I receive an e-mail with a job application that looks plausible. Since the attachment is cv-harward.txt, I consider ...	...this attachment to be low risk. This format is unusual. But it is unlikely that cyber criminals succeed in integrating malware here.	Attachment 4

D Additional Feedback Questions

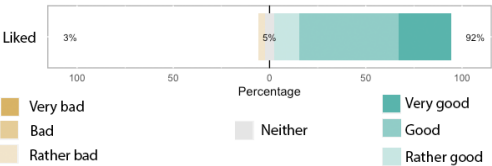


Fig. 9. Stacked bar chart for “How did you like the game card game in terms of design?”

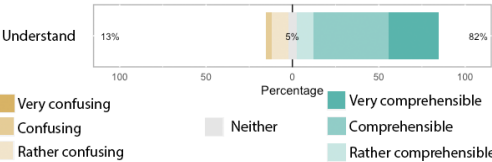


Fig. 10. Stacked bar chart for “How comprehensible are the instructions of the game?”

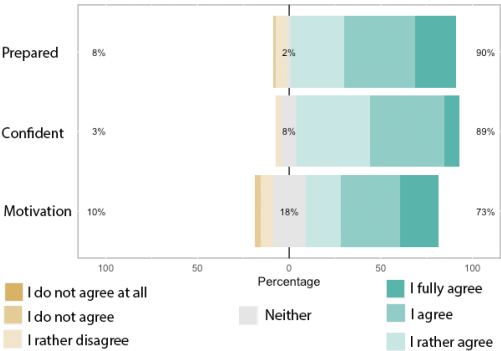


Fig. 11. Stacked bar chart for “Due to the solution after the 1st round, I was more motivated in the 2nd round”, “I feel better prepared to recognize possible phishing e-mails now that I have played the game.”, “I now feel confident in recognizing phishing e-mails.”

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